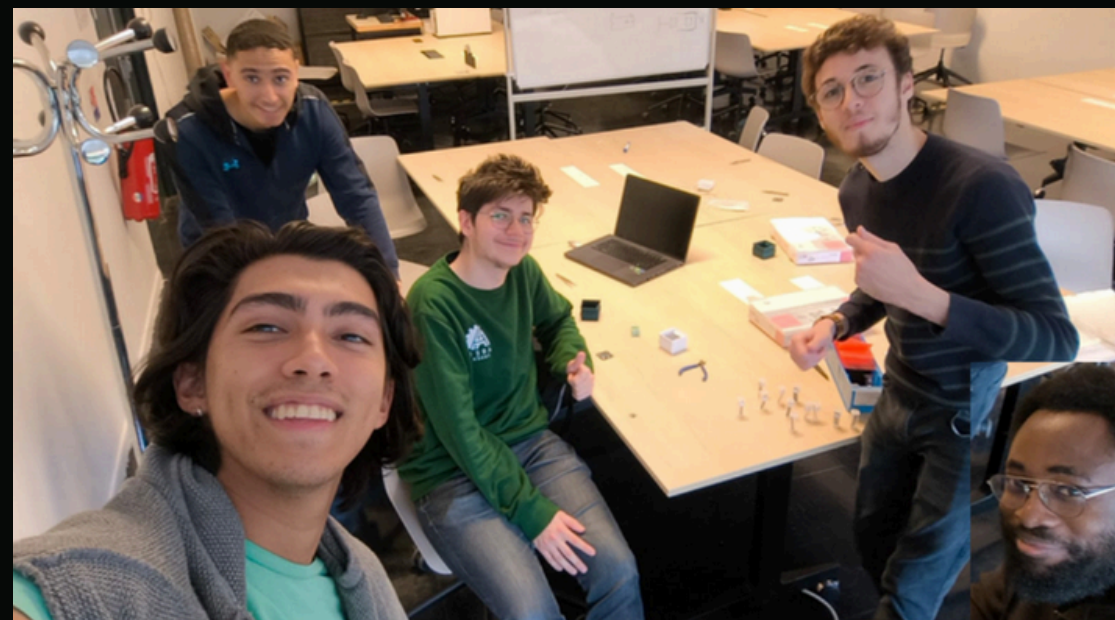




CanSat Vortex

Alexei Douillard • Ted Koyazande • Juan Pablo Barona Cifuentes •
Amr Taouis • Abdelmoughit Hajji Laamouri



TEAM VORTEX — ARES ENSEA

Mission objectives & Solutions

MISSION 1 • MANDATORY

Deployment & landing

Internal parachute with a fully mechanical deployment in flight, between 75 and 90 m above ground.

MISSION 2

Downsizing to 33 cl

The whole satellite fits a soda-can core
The 6 layers custom mainboard's dimensions are : 33mmx25mm

MISSION 3

Ground study

A spinning LiDAR maps the terrain into a 3D point cloud during descent.

MISSION 4

Onboard camera

An independent mini-camera records the landing on its own SD card and transmit the video feed live on a computer

FREE MISSION

Live telemetry

Real-time LoRa downlink of flight data to our ground station.

ADDITIONNAL INFRASTRUCTURE

Wind turbine

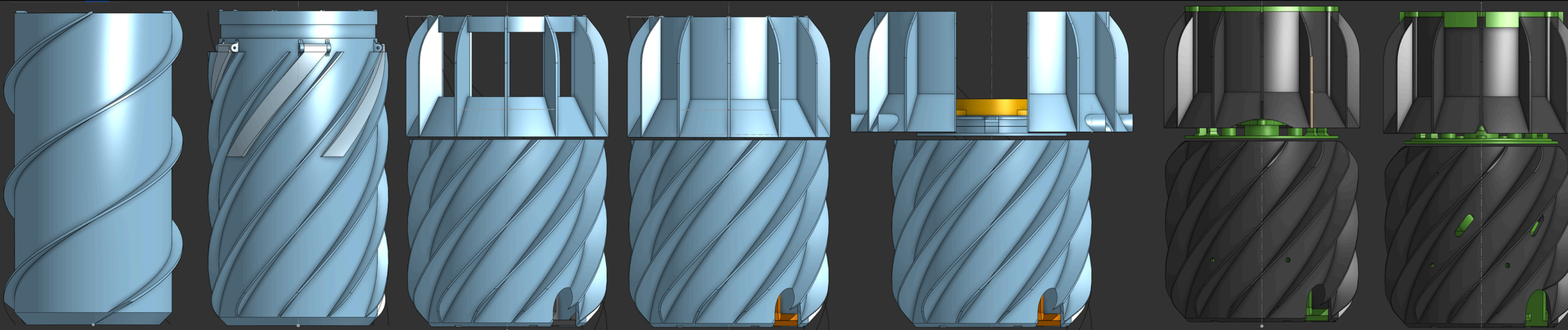
Custom wind turbine for interactive tests

CanSat component choice

SUBSYSTEM	REFERENCE	ROLE
Processor	STM32G431CBU6	170 MHz Cortex-M4 — fusion & flight logic
Barometer	BMP581	Altitude & descent rate — deployment trigger
IMU	BN0055 • 9-axis	On-chip fusion — orientation of every LiDAR shot
GNSS	u-blox SAM-M10Q	4-constellation position + trajectory
LiDAR	LightWare LW20/C	100 m laser — ground mapping @ 50 Hz
Radio	SX1276 LoRa	869.53 MHz downlink to ground station
Storage	microSD • FATFS	Full-rate telemetry + LiDAR logs, crash-safe
Power	2S LiPo 7.4 V • 2×500 mAh	Buck → 5 V, low-noise LDOs → 3.3 V rails

03 3D Model

Cansat Evolution



Wind tunnel validation

A homemade vertical tunnel driven by two 350 W counter-rotating brushless motors, airspeed is measured with an anemometer.

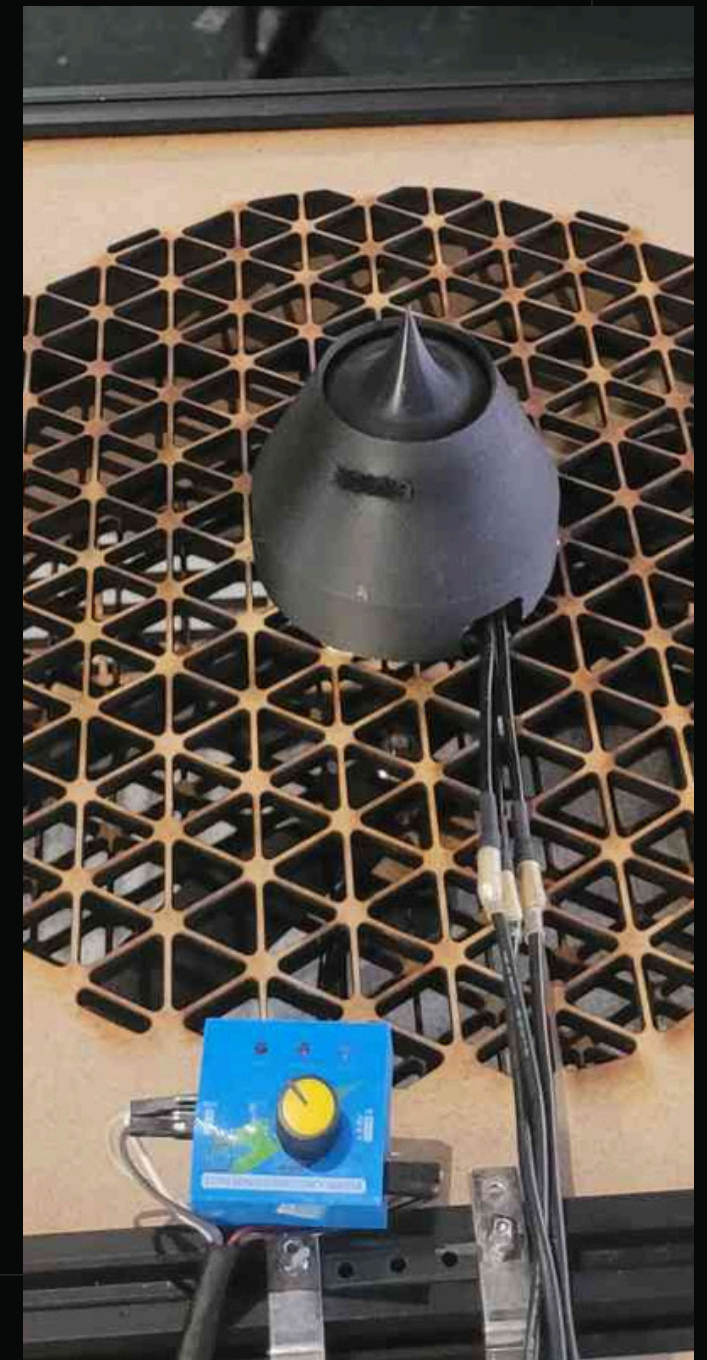
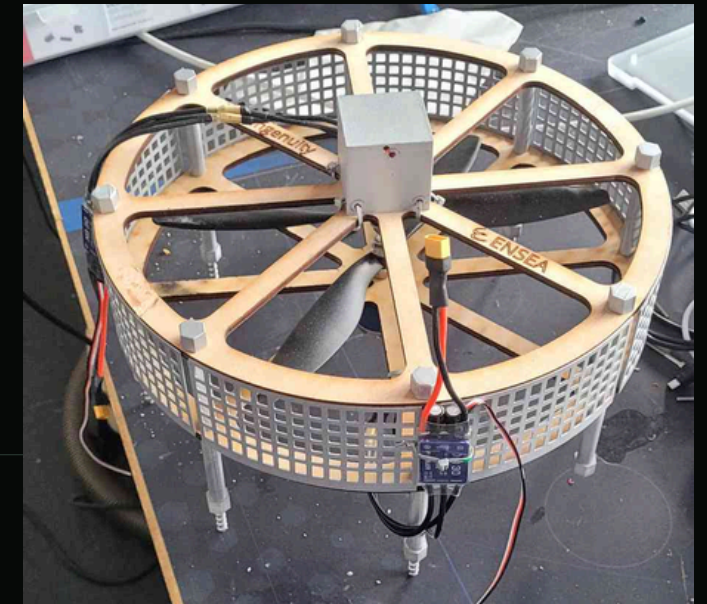
20 m/s

max airflow

2 × 350 W

counter-rotating drone motors

- Drogue chute drag characterisation
- Spin-stripe behaviour under real airflow
- Full deployment-chain experiments before drone drop tests

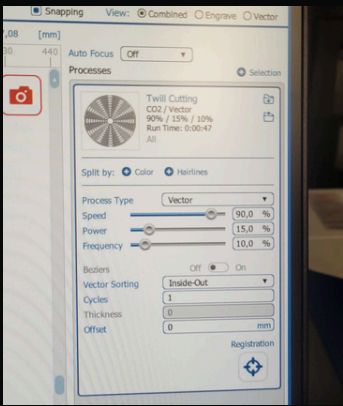
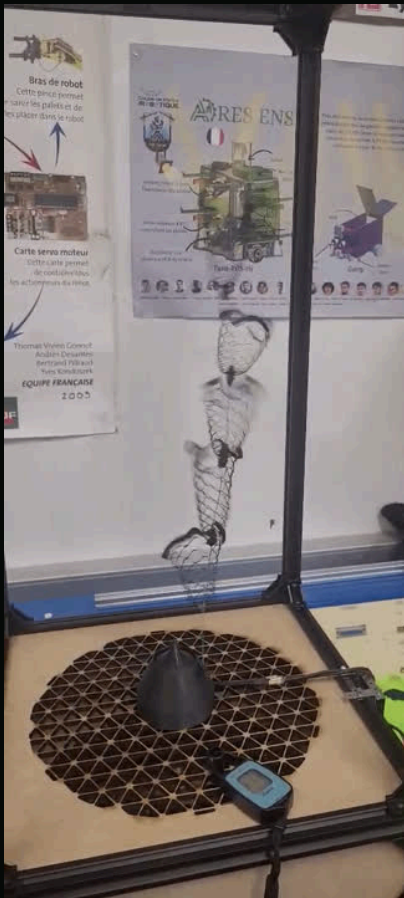
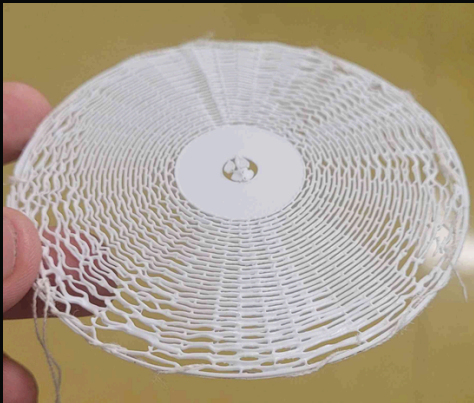


Parachute deployment system

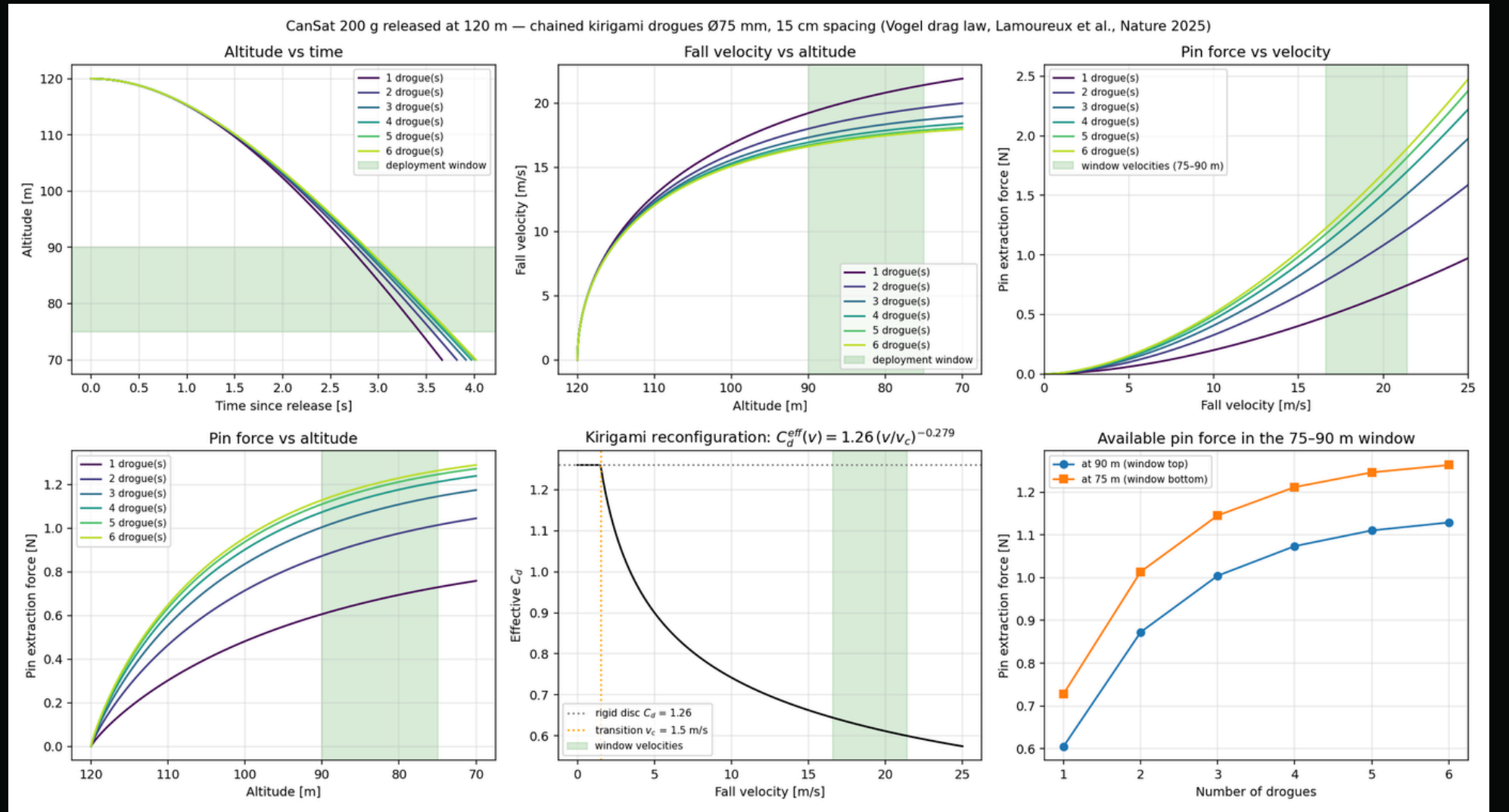
- 1
- Laser-cut Kirigami drogues from a K-Way Coat, chained to tune drag;
- 2
- Trigger release drogue drag pulls the pin; the spring-loaded triple fairing snaps open.
- 3
- Ø45cm chute out at 20m/s : Descent at 5m/s once deployed

0.6 N
extraction force

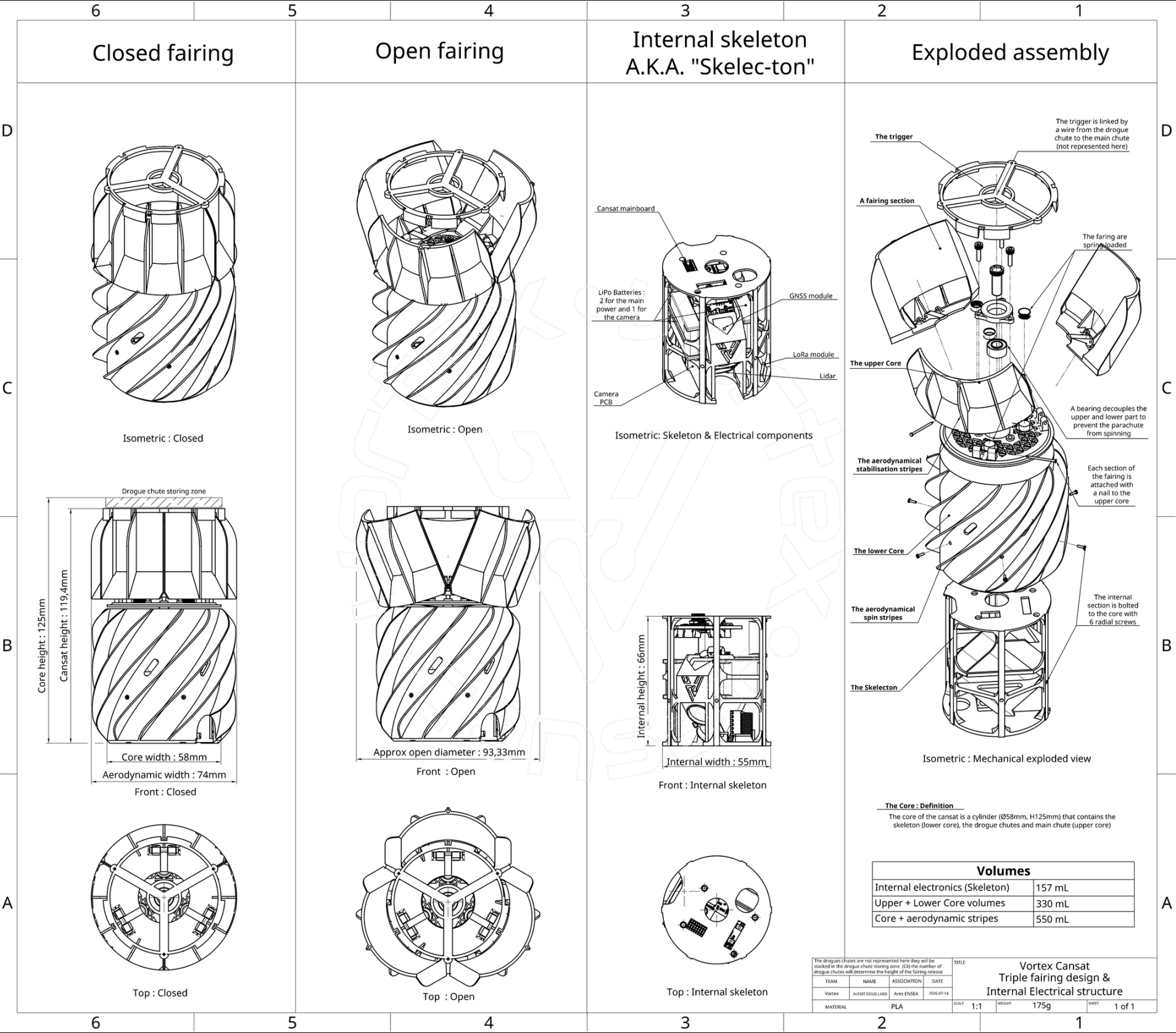
13 m/s
Slowest recorded drogue release speed
(3 drogues)



Expérience Pepe sur balance avec parachute bas peu pré				Expérience Pepe sur balance avec parachute et ressort coupés				Expérience pepe lesté sur balance avec parachute et ressorts coupés et goupille verte champfer 1,5 mm			
hole	intersect	random		hole	intersect	random					
54	40			84	39	26		56			
58	40			41	87	50		49	50	30	
45	62			46	99	13		48	61	54	
36	28			71	37	70		108	84	63	
48	47			38	73	45		49	40	33	
84	33			56	36	78		108	30	60	
60	25			92	32	57		33	36	37	
40	38			68	67	10		53	39	46	
70	74			105	58	50		40	93	88	
35	89			51	51	56		68	83	49	
100	28			46	42	46		49	48	46	
67	70			53	82	44		48	67	92	
65	38			74	78	34		34	28	14	
58,61538462	47,07692308			63,46153846	60,07692308	44,53846154		59	20	71	
23,53066739	23,53066739			20,88306884	22,38847002	19,91488298		57,28571429	52,23076923	52,53846154	
								24,30548107	23,53066739	22,42994793	



Vortex architecture



Ø58 × 125

core dimensions, mm

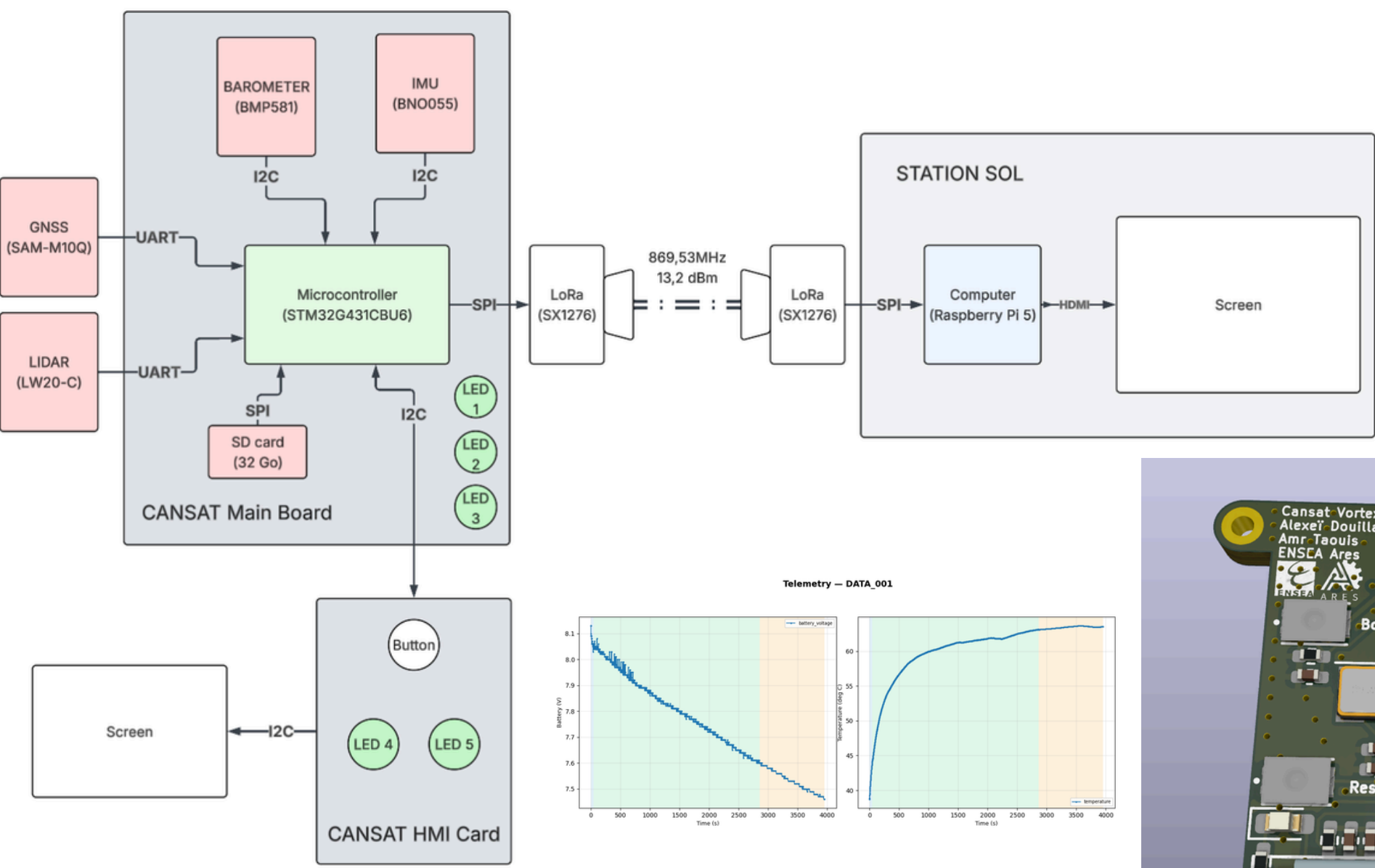
175 g

Total mass

- 01 **Upper core** — spring-loaded triple fairing storing main chute triggered by the drogue chutes
- 02 **Lower core** — full electronics skeleton spins to make the lidar scan possible
- 03 **Ball bearing** — decouples the chute so the body spins freely.
- 04 **Helical stripes** — passive fins that induce the LiDAR scan rotation.

Mainboard custom electronics

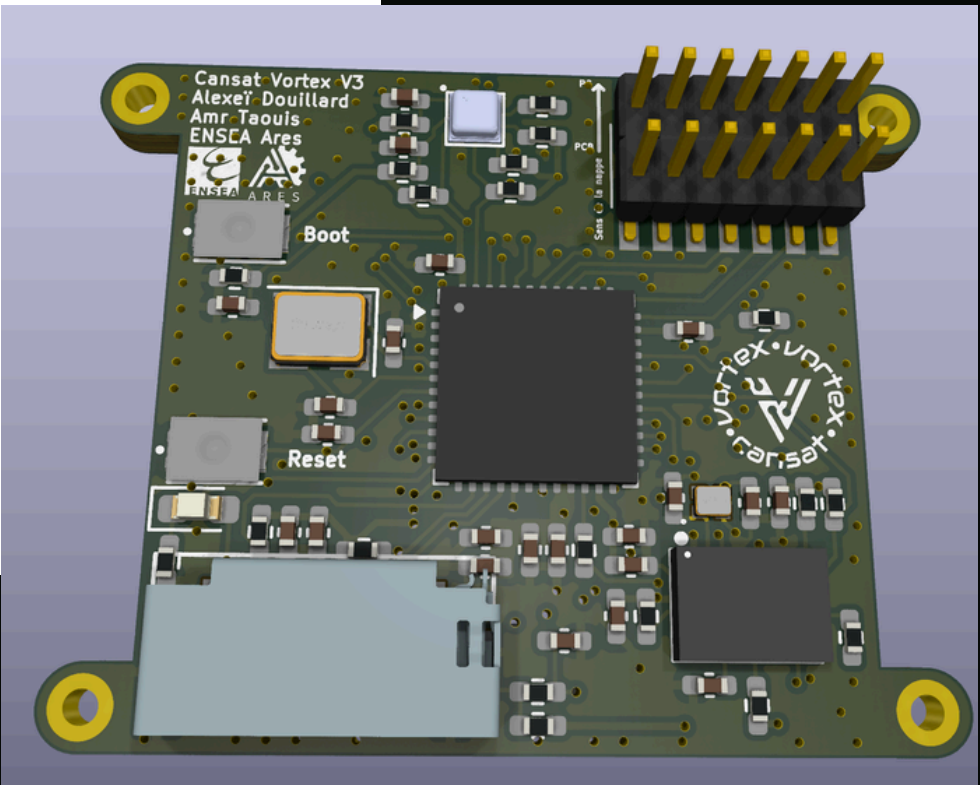
System Diagram



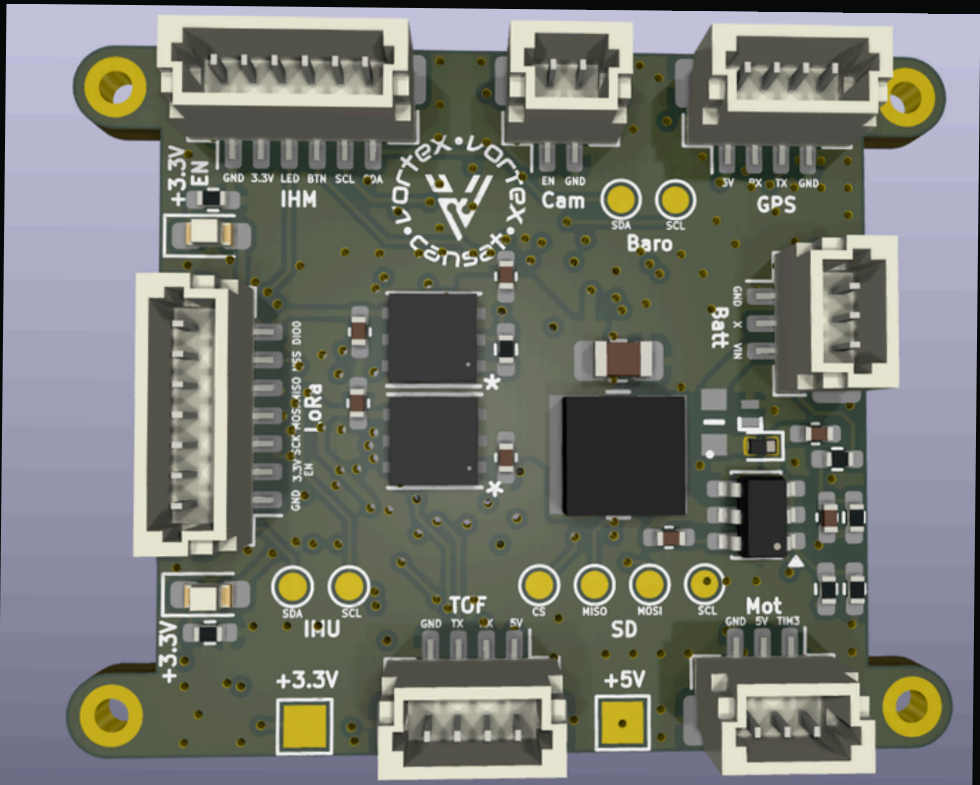
6-LAYER STACK-UP

SIG · GND · SIG · PWR · GND · SIG

- V1 Buck footprint + enable-line errors — found, hand-patched, board alive. 1st Prototype
- V2 Fixes rolled in; HMI status-screen and camera connectors added. Flight Version
- V3 Camera UART (not used), cleaner routing, added electrical protection. PCB Cursed by the solder paste



V3 FRONT — MCU · BARO · IMU · SD



V3 BACK — POWER · MODULE CONNECTORS

Onboard software

FREERTOS · 4 TASKS · QUEUE-BASED

TaskSensors

Baro · IMU · GNSS · LiDAR acquisition — high priority

TaskFSM

Flight state @ 20 Hz — builds the 10 Hz unified snapshot

TaskLoRa

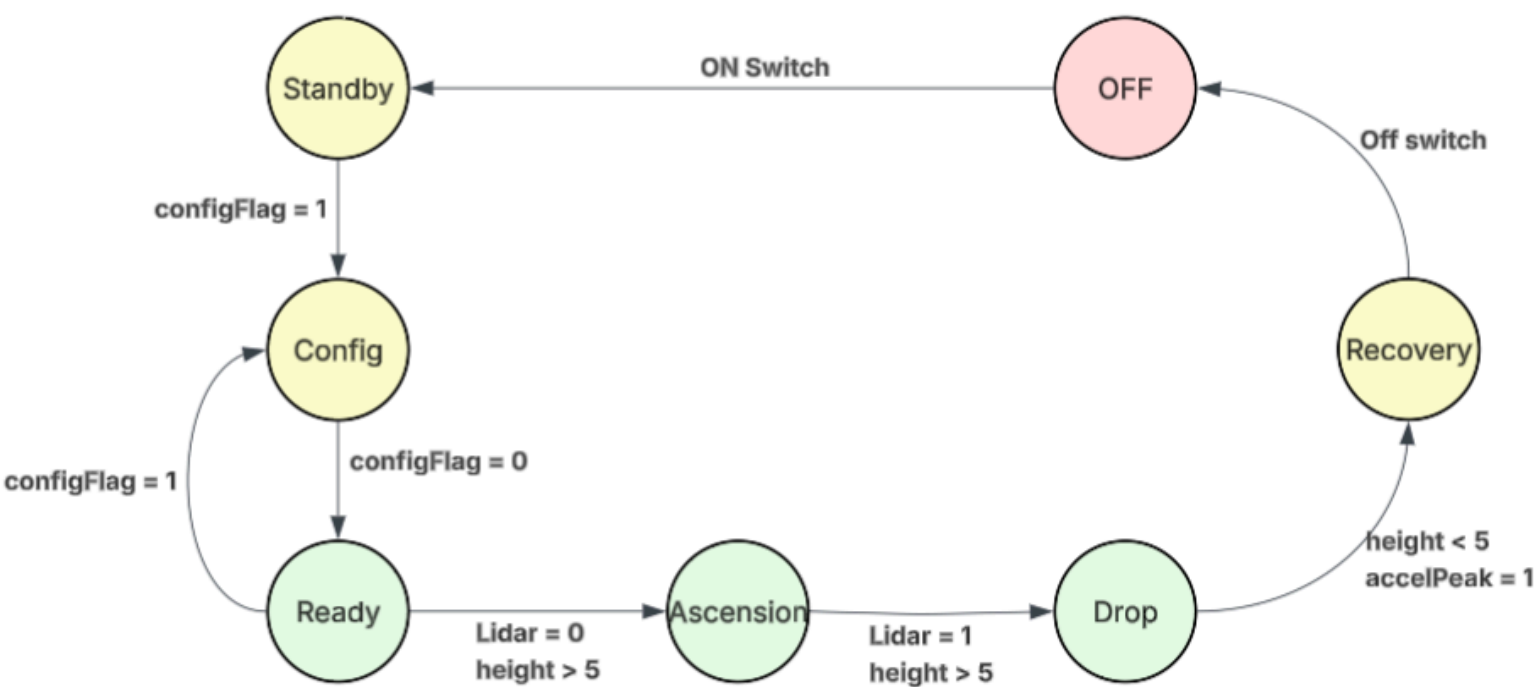
CRC-protected downlink of every snapshot

TaskSDCard

LiDAR @ 50 Hz + telemetry to CSV, batched f_sync — crash-safe

Fully autonomous mission no ground command needed after power-on, except for the camera.

FSM of the mission



LIDAR + BARO DRIVEN TRANSITIONS · 7 STATES

Ground station & HMI

869.53

MHz — EU ISM band

SF7 · CR4/8

125 kHz Bandwidth

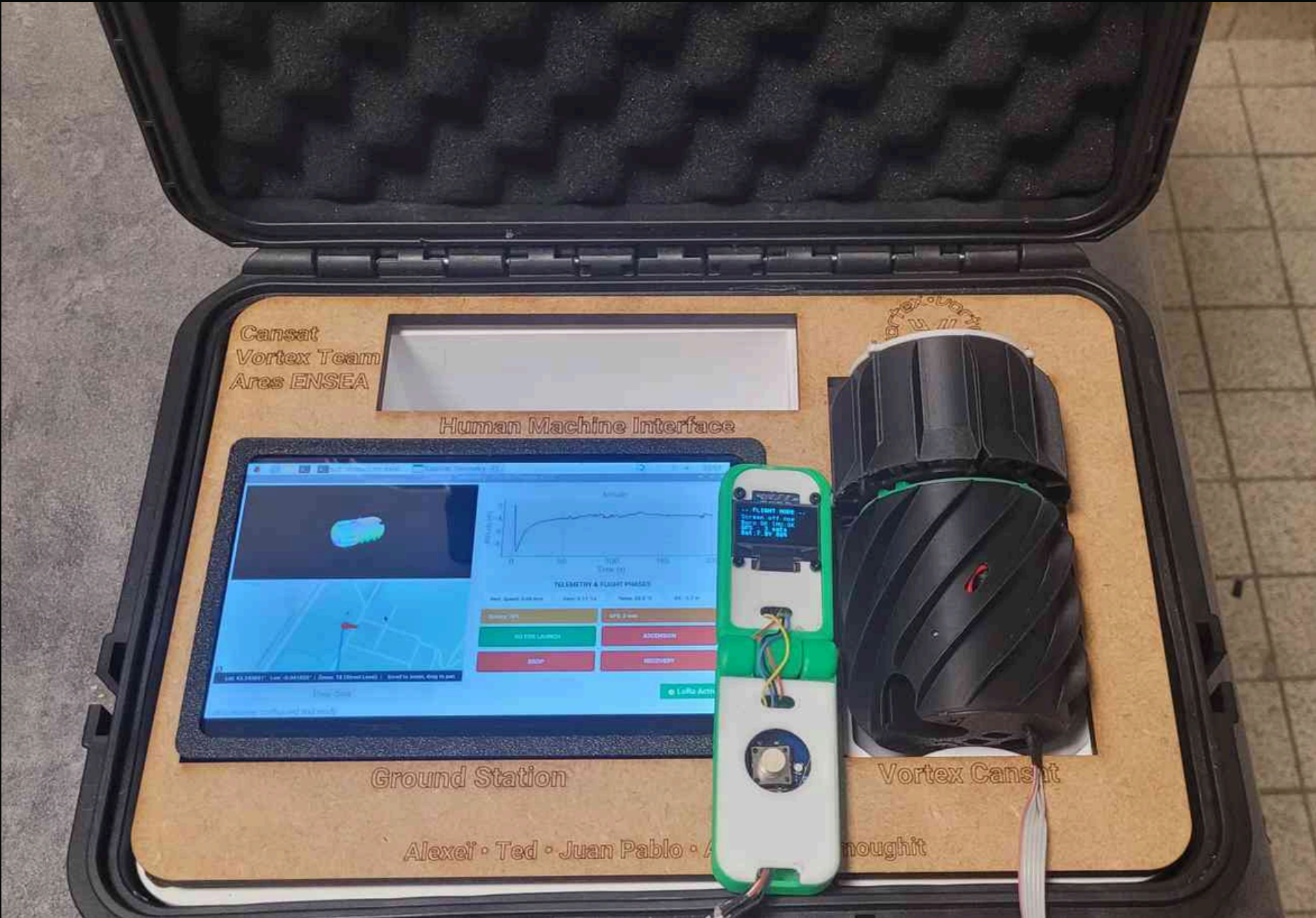
3D LIVE ATTITUDE

OSM GPS TRACKING

REAL-TIME PLOTS

FLIGHT-PHASE FLAGS

CSV LOGGING



PORTABLE MISSION-CONTROL CASE — RASPBERRY PI 5 + SX1276 LoRa module
Powered by 2 power banks

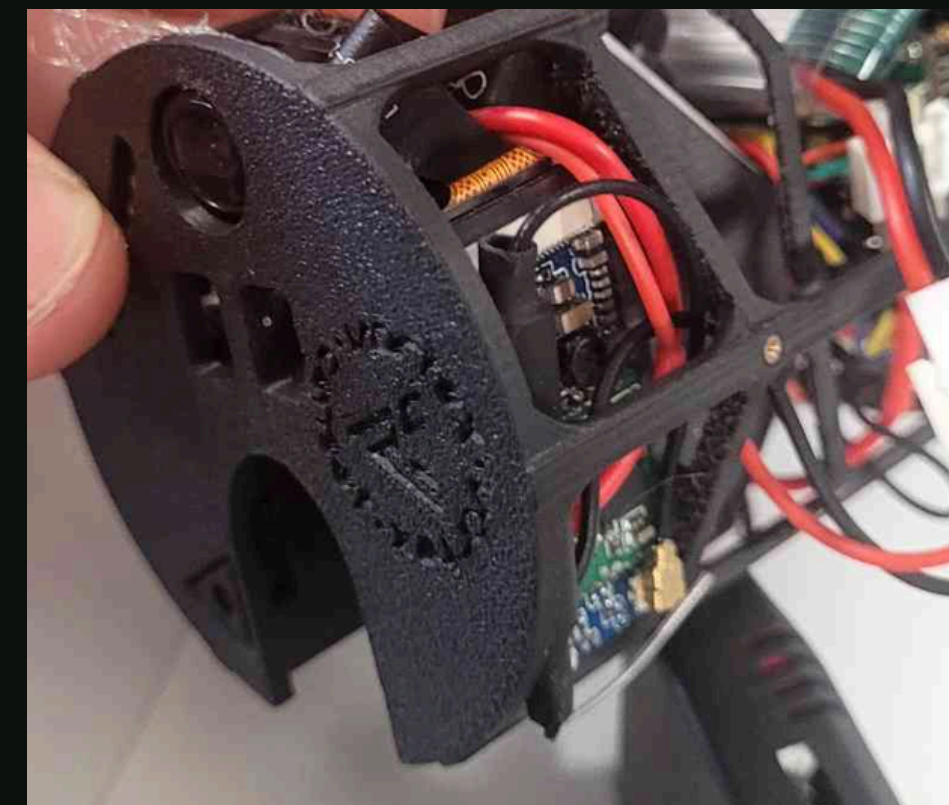
10 • Video recording

Video recording



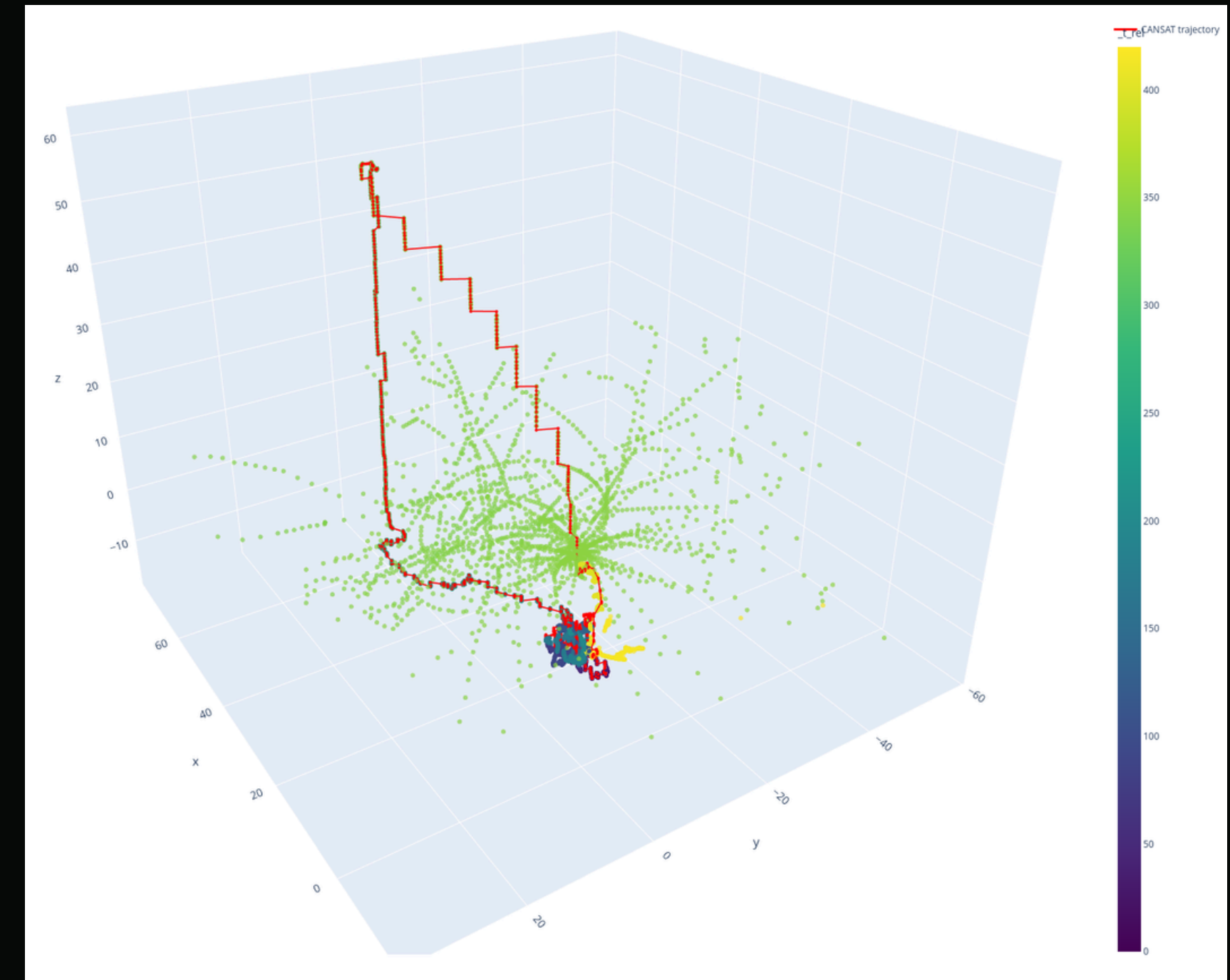
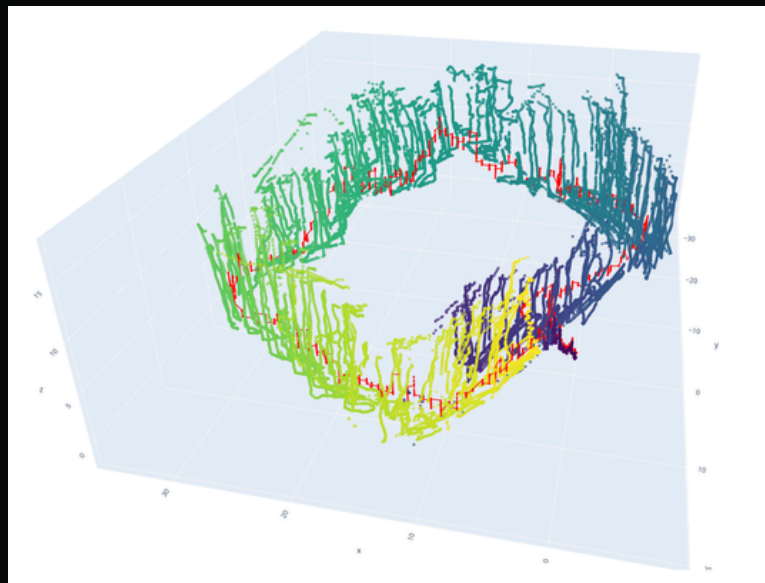
45m

max range of th wi-fi connexion



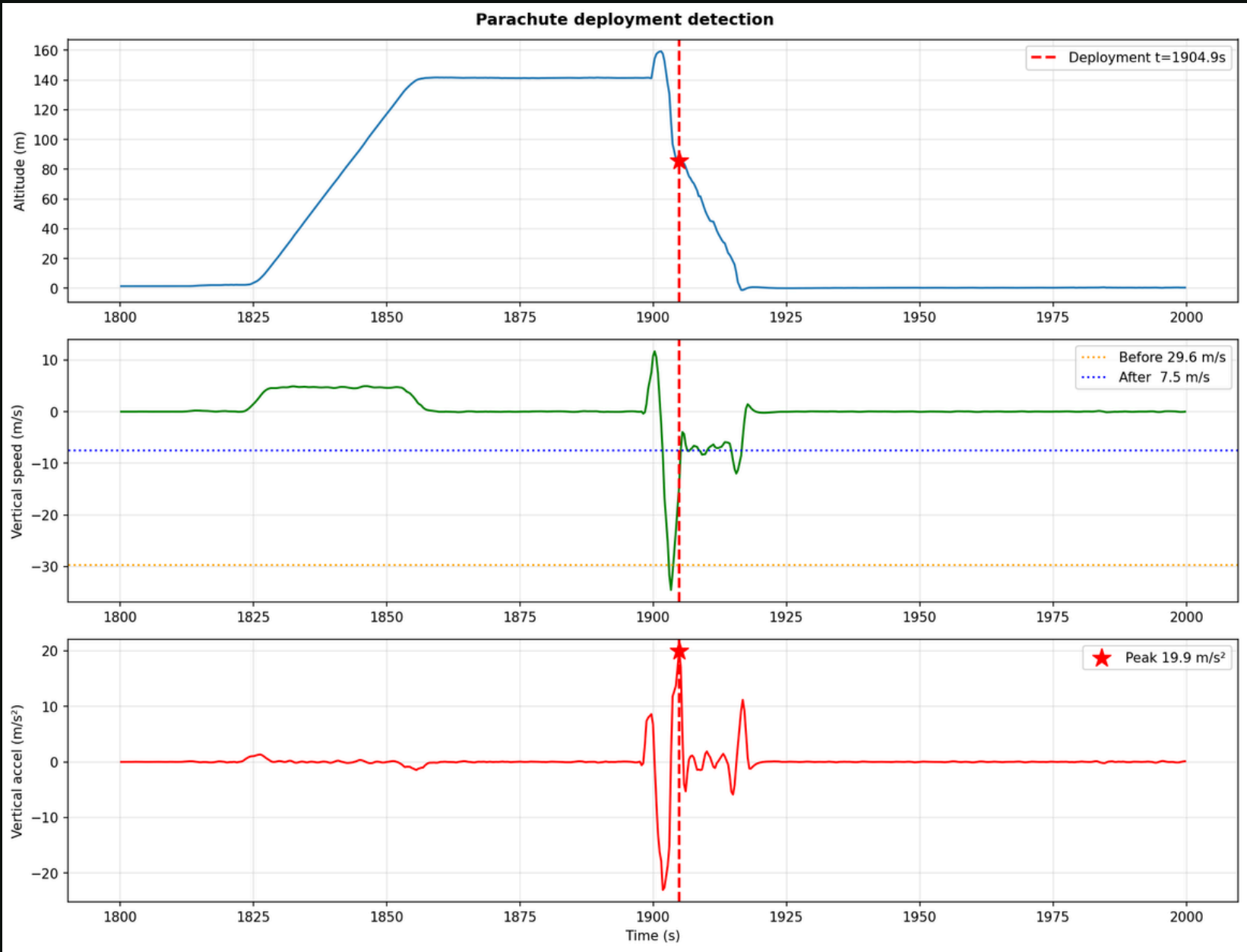
Ground study by spinning LiDAR

- 01 Helical stripes spin the body — **~1 rev / 15 m** of descent.
- 02 LW20/C samples ground distance at 388 **Hz**, logged to SD.
- 03 BNO055 quaternions orient each shot; baro + GNSS place it in space.
- 04 SLERP interpolation fills orientation between IMU bursts.
- 05 Our **cloudPoints.py** pipeline fuses everything → XYZ cloud → CloudCompare.



FIELD DROP TESTS — DRONE RELEASE MECHANISM

Flight results — Main mission



Parachute Deployment

MISSION 3 — SPIRAL SCAN → XYZ CLOUD → CLOUDCOMPARE



14 • DROP DAY

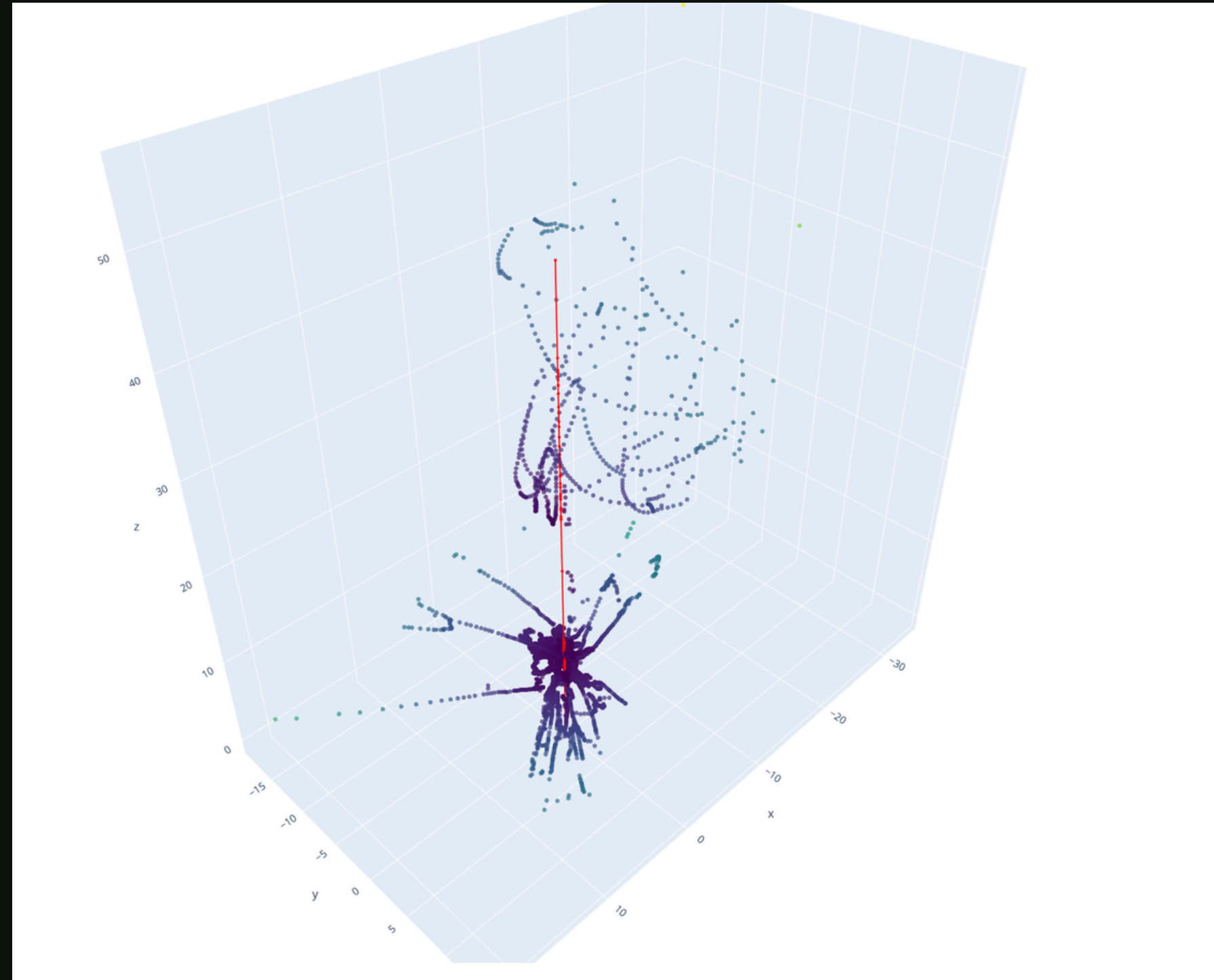
Flight results — point cloud & landing



MISSION 4 — INDEPENDENT ONBOARD CAMERA

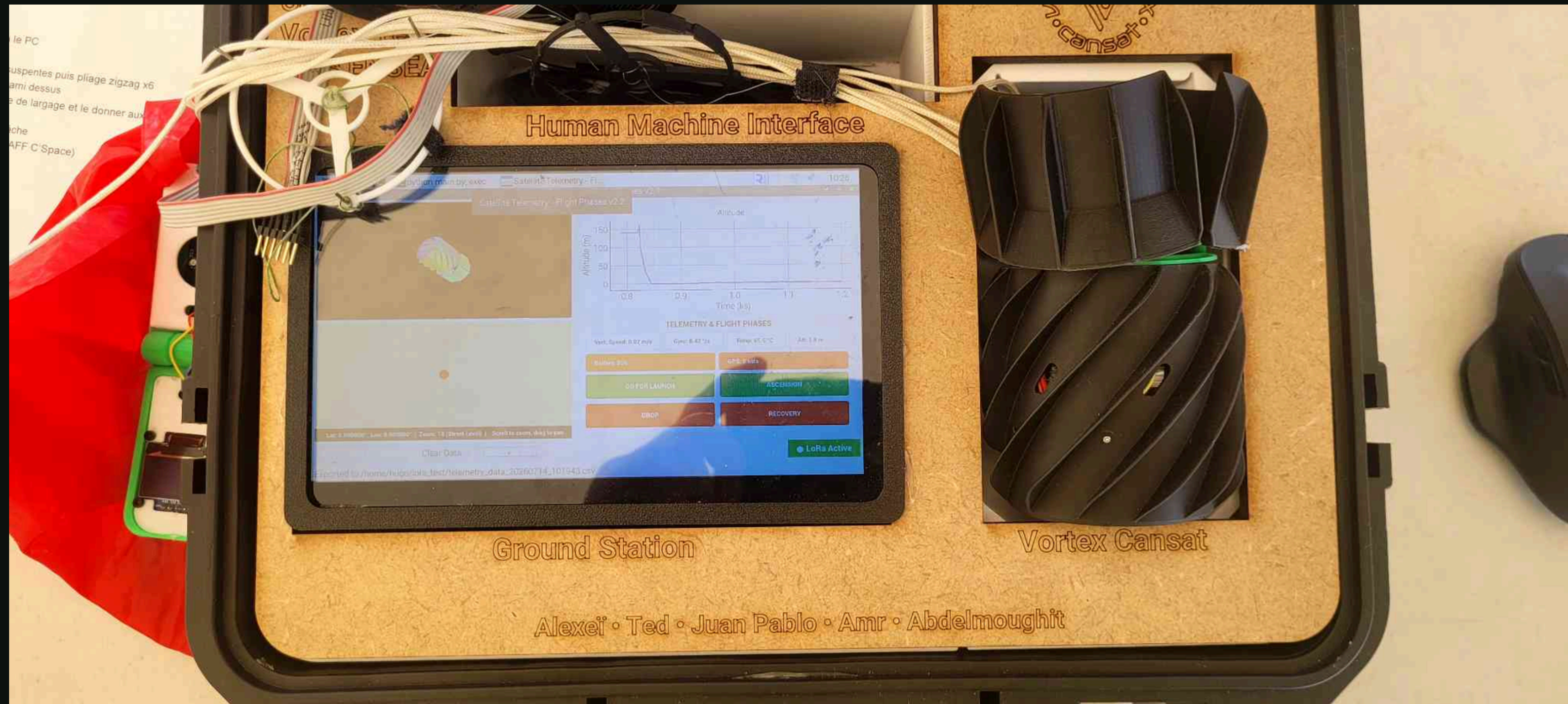
MISSION 3 — SPIRAL SCAN → XYZ CLOUD → CLOUDCOMPARE

Flight results — point cloud & landing



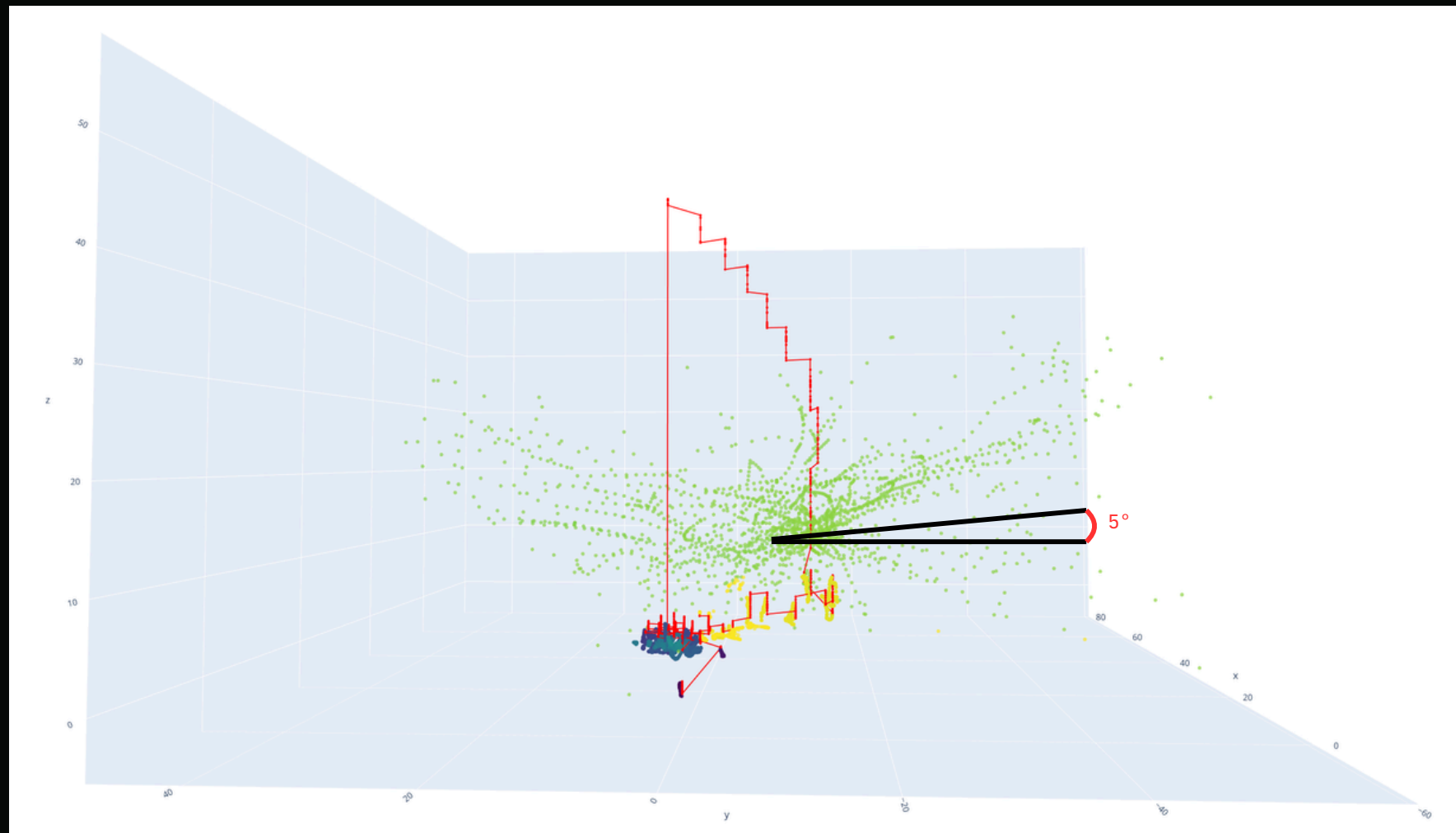
MISSION 3 — SPIRAL SCAN → XYZ CLOUD → CLOUDCOMPARE

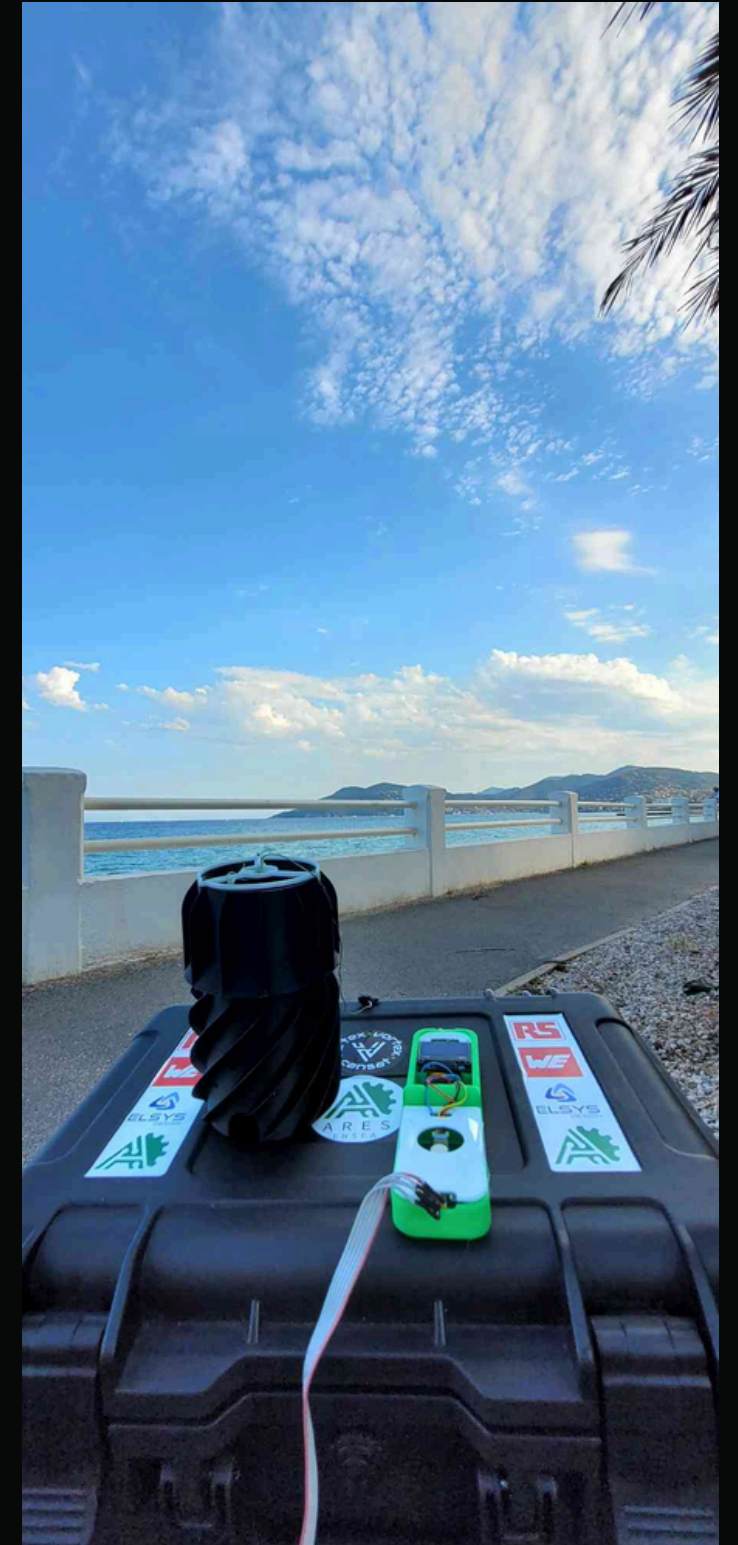
Flight results — telemetry



Known issues :

Tarbes : $43^{\circ} 14' 0''$ N / $0^{\circ} 5' 0''$ E
Paris : $48^{\circ} 52' 0''$ N / $2^{\circ} 20' 0''$ E
5° latitude difference for the IMU





Thank you.

Questions welcome

Hardware, firmware, ground station and analysis pipeline
are open in the team repository.





Identity Card

Country:	France
City:	Cergy
1st C'space:	2024
Previous Launch:	2025 Les Crackitos
Current project:	2026 Vortex

About Us

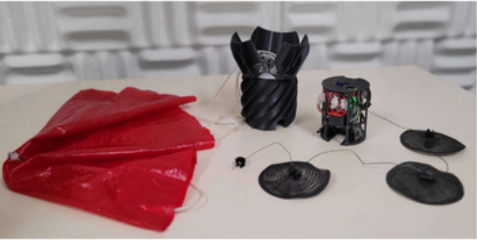
Ares is the robotics association of ENSEA (École nationale supérieure de l'électronique et de ses applications).
The Vortex team built on last year's expertise and created a wind turbine with conter rotative drone propellers to test the rotation and chute deployment mechanism.



Team Member	Role
Alexeï DOUILLARD	3D Modeling, PCB, Satellite integration, Head of project
Juan Pablo BARONA CIFUENTES	Embedded systems, firmware
Ted KOYAZANDE	Ground Station, Communications
Amr TAOUIS	PCB Design and testing
Abdelmoughit HAJJI LAAMOURI	Code

Cansat 2026 missions

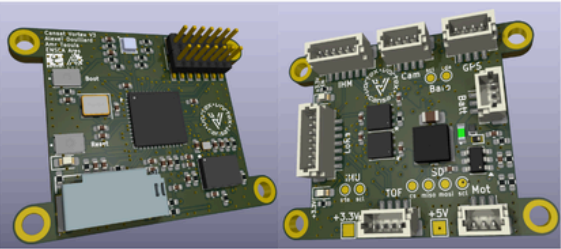
- **Downsizing** : Integrating the core system within the volume of a 33cl soda can
- **Deployment of a fully contained parachute deployed between 75 and 90m** : Using an extractor parachute (drogue chute) inspired by kirigami calibrated for the 15m window.
- **Ground Mapping** : Generating a 3D map of the terrain through rotation using a LiDAR, GPS, barometer, and an inertial measurement unit.
- **Onboard Camera** : Recording the landing using a drone camera.
- **Bonus mission** : telemetry transmission to a ground station



Vortex Cansat C'SPACE 2026

Hardware : Electronics

Design of a 6-layer PCB integrating an STM32 microcontroller, SD card, barometer, and 9 axis IMU, with dedicated connectivity for GPS, LoRa, camera, and LiDAR modules.

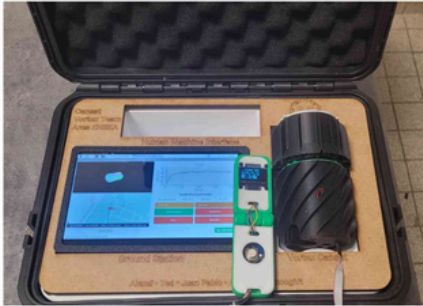


Software & Telemetry

Development of an embedded software architecture using FreeRTOS (C programming language). It manages sensor fusion, data logging to the SD card, and real-time flight data transmission via a LoRa module.

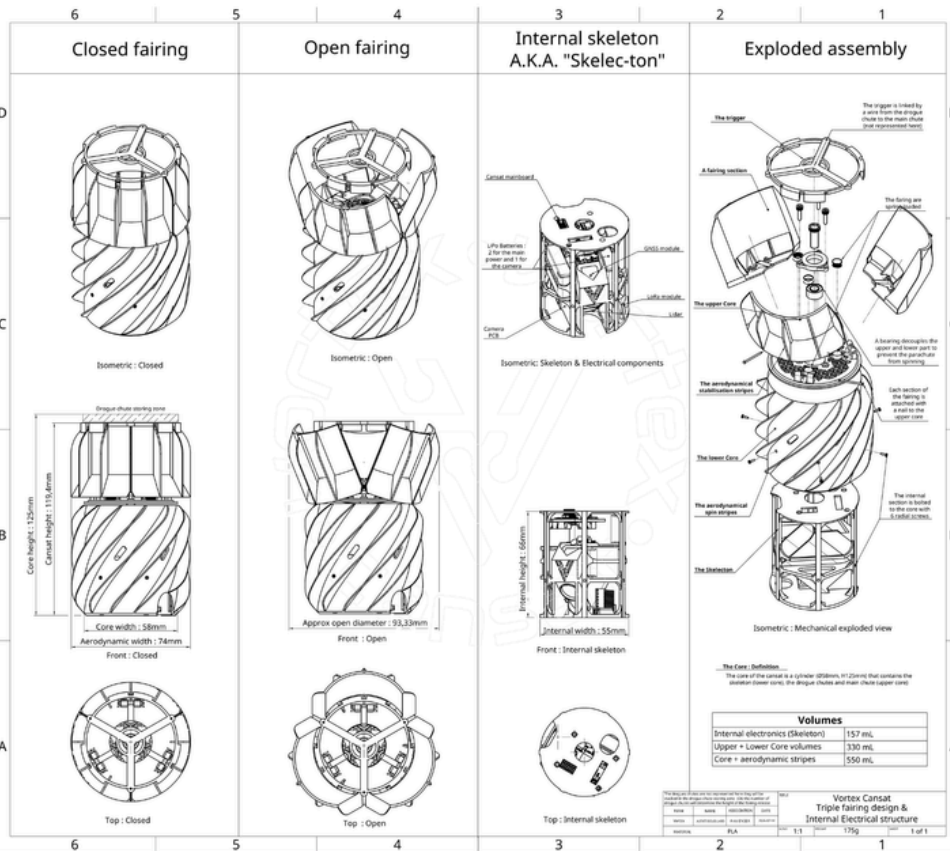
Ground Station & HMI

Creation of an autonomous ground station powered by a Raspberry Pi 5. It features a dashboard with real-time 3D orientation modeling of the CanSat, GPS tracking on a map, and live telemetry graphs. We also built a plug in Human Machine Interface for the preflight checks.



Mechanics & Aerodynamics

The 3D design of the satellite has a helical body engineered to induce rotation during descent. It enables the lidar to scan the floor and create a spiral topographical map. The electronics is secured in an internal skeleton. To deploy the main Ø45cm parachute, laser cut kirigami drogue parachutes pull on a trigger. To ensure deployment at a precise speed we tuned the trigger to adjust friction and chained drogues chutes to get enough force.



Test Flights

1-80m drop Testing parachute for the future V3 test
2-120 ascension Testing FSM and behaviour of the V2
3-130m drop with 2 drogues chutes
4-Test camera Wi-Fi range
5-50m Test Lidar Scan V3
6-60m test triple parachute
7-155m Real C'Space Launch
8-60m Lidar Scan V2
9-50m Drop Rock Sat for the open doors presentation
10- 135m Drop